JEPA-Contract for Sturm-Constrained Learning of One-Dimensional Schrödinger Eigenstates

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Learned surrogates for quantum eigenvalue problems must approximate Hamiltonian-dependent eigenstates while preserving known a priori spectral structure. In this paper, we introduce a physics-structured operator-to-state surrogate in which a joint-embedding predictive architecture learns the Hamiltonian-dependent state representation, while a phase-based decoder enforces Dirichlet boundary conditions and the Sturm nodal class exactly. Across 783 held-out Hamiltonians spanning 27 potential families, the direct predictions achieve a mean fidelity of 0.9786. A reduced Rayleigh–Ritz rotation within the learned span increases the mean fidelity to 0.9942 and lowers the mean Schrödinger residual at every spectral index, while its effect on state-resolved nodal identity remains strongly state-dependent. A controlled ablation without the physical structure achieves comparable aggregate fidelity but loses nodal class, spectral ordering, and residual-based reliability. These results show that structural admissibility, operator consistency, and variational quality are distinct properties of learned eigenstates. Inference requires no reference eigenvalues or eigenvectors.

Solving the Schrödinger eigenvalue problem repeatedly across families of Hamiltonians remains a basic computational task in quantum physics. Parameter sweeps, inverse problems, and large ensembles of potentials motivate learned representations that complement conventional numerical solvers. Neural quantum-state approaches optimize variational wavefunctions for a given Hamiltonian, often incorporating physical symmetries [1–5], whereas supervised surrogates learn to predict observables or wavefunctions from numerically generated solutions [6–8]. Physics-informed neural networks incorporate governing equations and boundary conditions into the optimization objective, with applications ranging from incompressible flows to wave propagation [9, 10]. Neural operators provide a complementary approach by learning mappings between function spaces and have been developed as surrogates for parameterized physical systems, including nonequilibrium flows and Maxwell equations [11, 12]. These approaches raise a common question: which properties of a predicted quantum state can be guaranteed structurally, and which require consistency with the supplied operator?

For one-dimensional bound states, spectral theory provides a particularly sharp distinction. A regular Sturm–Liouville operator with Dirichlet boundary conditions has an ordered spectrum $E_0 < E_1 < \cdots$, and its nth eigenfunction contains exactly n interior zeros [13, 14]. The state index therefore fixes a nodal class, not merely a position in the energy spectrum. A regression model can approximate a wavefunction with high overlap while adding spurious zeros or missing physical ones. Conversely, a function with the correct number of nodes need not be an eigenfunction of the Hamiltonian under consideration.

This work introduces a physics-structured operator-to-state surrogate that makes this distinction explicit. A joint-embedding predictive architecture (JEPA) maps each Hamiltonian to a latent eigenstate representation, while a phase-based decoder enforces its Sturm nodal class. Exact enforcement of the Sturm nodal class is shown to guarantee structural admissibility but not operator consistency. Learned topology-preserving states can span an accurate variational eigenspace, while variational optimization need not preserve their state-resolved nodal identities. The central comparison is therefore between the direct, structurally constrained states and the states obtained by a Rayleigh–Ritz rotation within their span.

The eigenvalue problem $\hat{H}[V]\psi_n = E_n\psi_n$ is considered on $x \in [a, b]$, with $\hat{H}[V] = -\kappa\partial_x^2 + V(x)$, $\kappa = \hbar^2/(2m)$, and homogeneous Dirichlet boundary conditions. The surrogate approximates

$$[V(x), n] \longmapsto \{\widehat{E}_n, \widehat{\psi}_n(x)\}. \tag{1}$$

This operator-to-state formulation is related to neural-operator surrogates, but here the output is additionally conditioned on the spectral index and constrained by the known Sturm structure of the target state.

Finite-element (FEM) eigensolutions provide training targets. A solution autoencoder first constructs a latent representation of these states; a context encoder and state-conditioned predictor then learn to reach this representation from the potential and domain information alone. The context–predictor–target organization follows the JEPA paradigm and its I-JEPA realization [15, 16]. Latent alignment is followed by progressive physical supervision. The target-encoder update, reconstruction tests, and training details are given in the End Matter (EM).

The complete learning and inference workflow is summarized in Fig. 1. The architecture separates three roles: the Hamiltonian-dependent representation learned by the context–predictor branch, the spectral structure imposed by the constrained decoder, and the variational correction performed by Rayleigh–Ritz within the learned span. Training follows a curriculum in which a latent solution manifold is first established through solution-autoencoder pretraining. Operator-to-state latent alignment is then learned, the supervised spectral subspace is expanded from 3 to 6 and finally 11 states, and physical consistency terms are introduced progressively before joint

fine-tuning. The corresponding optimization stages and their training objectives are shown in Fig. 1(b)–(c).

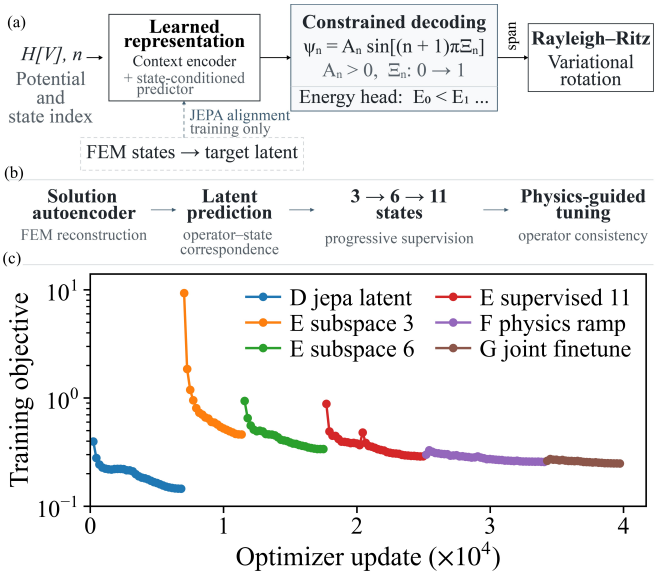


FIG. 1. Architecture and curriculum training of the physics-structured operator-to-state surrogate. (a) Operator-to-state map. The supplied Hamiltonian $H[V]$ and state index n are encoded into a learned latent representation and decoded through a physics-constrained branch that imposes Dirichlet boundaries, the continuous Sturm nodal class, and spectral ordering. During training, FEM eigenstates provide the target latent representation through JEPA alignment; this target branch is removed at inference. The decoded states span the subspace used by the reduced Rayleigh–Ritz rotation, which improves variational consistency without guaranteeing preservation of state-resolved nodal identity. (b) Curriculum used to train the model: solution-autoencoder pretraining, latent operator-to-state prediction, progressive supervision over $3 \to 6 \to 11$ states, and final physics-guided tuning. (c) Training objective along the corresponding curriculum stages. Each color denotes a distinct optimization stage; discontinuities between stages reflect changes in the active objective and supervised state set rather than a single continuous loss function.

The decoder separates nodal class from spatial shape. In the coordinate $t = (x-a)/(b-a)$, a positive phase density $q_n(t) = \mathrm{softplus}[\eta_n(t)] + \varepsilon$, with $\varepsilon > 0$, and a positive amplitude $A_n(t)$ define

$$\Xi_n(t) = \frac{\int_0^t q_n(s)\,\mathrm{d}s}{\int_0^1 q_n(s)\,\mathrm{d}s}, \qquad \widehat{\psi}_n(t) = A_n(t)\sin[(n+1)\pi\Xi_n(t)]. \tag{2}$$

Because Ξ_n increases strictly from zero to one, the phase crosses each of $\pi, \ldots, n\pi$ exactly once in the interior. The continuous output thus has exactly n interior nodes and vanishes at both endpoints. The amplitude, node positions, localization, and relative lobe weights remain learned quantities. This exact nodal statement concerns the continuous decoder; resolving every zero on a finite grid is a separate numerical question.

The energy branch likewise imposes ordering through positive adjacent gaps, while learning quantitative corrections to an operator-only auxiliary spectrum in a fixed sine basis. At inference, neither FEM eigenvalues nor eigenvectors nor the solution encoder are used. The reduced auxiliary diagonalization depends only on $V(x)$ and is not a full FEM solve.

Structural constraints do not enforce the eigenvalue equation or mutual orthogonality. Let $\widehat{\mathbf{c}}_n$ represent the decoded state in the FEM basis, with mass matrix $\mathbf{M}$ and Hamiltonian matrix $\mathbf{H}$, both assembled from the supplied operator. Operator consistency is tested through the weak residual

$$\mathbf{r}_n = \mathbf{H}\widehat{\mathbf{c}}_n - \widehat{E}_n\mathbf{M}\widehat{\mathbf{c}}_n. \tag{3}$$

For a complementary multi-state test, the predicted vectors are orthonormalized in the $\mathbf{M}$ metric to form the columns of $\mathbf{Q}$. The reduced problem is

$$\mathbf{H}^{\mathrm{proj}} = \mathbf{Q}^{\dagger}\mathbf{H}\mathbf{Q}, \qquad \mathbf{H}^{\mathrm{proj}}\mathbf{u}_n = E_n^{\mathrm{RR}}\mathbf{u}_n. \tag{4}$$

The Ritz state has coefficients $\mathbf{Q}\mathbf{u}_n$. Thus the network supplies a trial subspace, while the variational calculation within it is performed explicitly. Linear combinations in this subspace need not remain in the nodal classes assigned to the original decoder outputs.

Figure 2 summarizes the central tension between operator consistency and state-resolved topology for the 783 in-distribution (ID) test Hamiltonians, each containing 11 states ($n = 0, \ldots, 10$). The complete dataset contains 5184 Hamiltonians from 27 potential families; complete operators, rather than individual states, are assigned to disjoint training, validation, and test partitions (EM). Rayleigh–Ritz lowers the mean normalized Schrödinger residual at every spectral index, showing that rotation within the learned trial subspace produces states that are more consistent with the supplied Hamiltonian. The reduction is systematic at the level of individual states, 86–99% of which are improved at each n, although it is small compared with the spread across Hamiltonians; the mean paired reduction grows from approximately 0.01 for the low-lying states to 0.034 at $n = 10$. Its effect on the numerically resolved topology, however, is strongly state dependent.

The same variational improvement is observed in complementary diagnostics. The mean fidelity increases from 0.9786 to 0.9942, the mean relative H^1 error decreases from approximately 0.100 to 0.061, the mean maximum Gram-matrix deviation from the identity decreases from approximately 0.124 to 2×10^{-6}, and the mean independent residual decreases from 0.108 to 0.092. At the same time, Fig. 2(b) shows that the effect on numerically resolved topology is not uniform. Rayleigh–Ritz reduces the resolved-topology fraction for the low-lying states $n = 0$–3, from 1.00 to 0.82 for $n = 0$ and from 0.85 to 0.75 for $n = 3$, whereas the higher states $n \geq 6$ exhibit improved preservation, reaching 0.995–1.00 for $n = 9$ and 10; the intermediate states $n = 4$ and 5 change in opposite directions. For the direct route, the resolved-topology fraction remains below unity for $n \geq 1$ despite the exact continuous Sturm class. These failures are concentrated in the half-line and radial families (`Half_harmonic`, `Kratzer`, `Radial_coulomb`, and `Radial_oscillator`), where the persistent-node criterion is

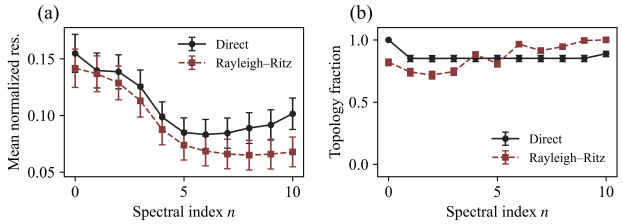


FIG. 2. Operator consistency and resolved topology on the ID test set. (a) Mean normalized Schrödinger residual as a function of spectral index n for the direct predictions and after Rayleigh–Ritz rotation. The variational rotation lowers the mean residual at every spectral index, with the largest reductions for the highest states. (b) Fraction of states satisfying the adopted numerically resolved topology criterion. In contrast to the residual reduction, the topological effect of Rayleigh–Ritz is state dependent: the resolved-topology fraction decreases for the low-lying states $n = 0$–3 and for $n = 5$, and increases for $n = 4$ and $n \geq 6$. Error bars denote 95% confidence intervals (Wilson intervals in panel b).

TABLE I. Selected eigenstates of a representative ID double-well test Hamiltonian, chosen as the one with the median mean direct fidelity among the 29 double-well test Hamiltonians. For each spectral index n, the table compares the FEM reference energy E_n^{FEM} with the direct surrogate prediction $\widehat{E}_n$, together with the state fidelity $\mathcal{F}_n$ and the relative energy error δE_n.

n	E_n^{FEM}	$\widehat{E}_n$	$\mathcal{F}_n$	δE_n (%)
0	0.195887	0.195417	0.997983	0.240
3	1.095722	1.079670	0.999305	1.465
7	3.178515	3.243393	0.999593	2.041
10	5.159921	5.211432	0.999692	0.998

not met near the singular endpoint. Variational improvement therefore does not imply either systematic preservation or systematic destruction of state-resolved nodal identity.

The distinction can be seen directly in the spatial predictions. Figure 3 illustrates this for a representative double-well Hamiltonian, with selected FEM and predicted states shifted to their respective energies. Table I lists the corresponding energies and fidelities for a representative double-well Hamiltonian of the ID test set. The decoder fixes the number of nodes across the spectrum, while agreement in their positions and in the lobe amplitudes reflects successful learning of the supplied potential. Across the ensemble, the gap between the mean (0.9786) and median (0.9989) direct fidelity reveals a low-fidelity tail rather than uniform reconstruction quality. A correct nodal class therefore remains compatible with substantial spatial error.

Numerically resolved zeros must also be distinguished from physically meaningful nodes. Figure 4(a)–(b) shows an $n = 5$ state of an ID single-barrier Hamiltonian, a case in which the direct output has the correct raw nodal class while the Rayleigh–Ritz state does not, even though the rotation increases the fidelity and lowers the residual. Rayleigh–Ritz increases the fidelity from 0.9905 to 0.9986 and lowers the independent residual from 0.046 to 0.039; both routes place the

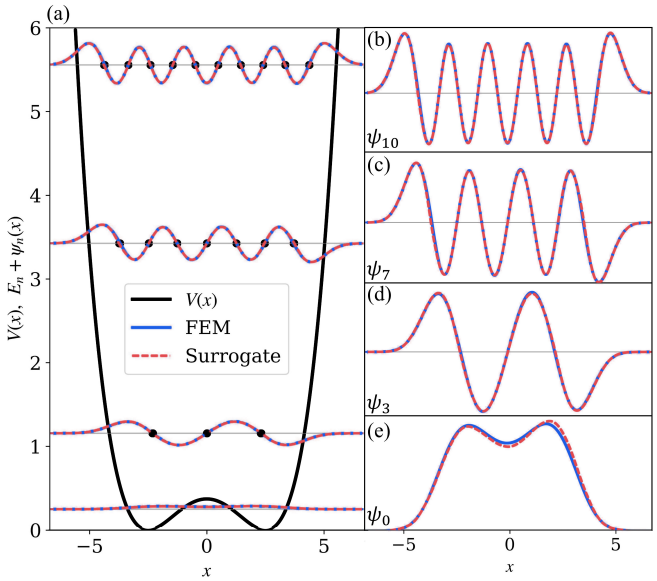


FIG. 3. Illustrative direct spectral reconstruction for an ID double well. (a) Potential, selected energies, and energy-shifted FEM and predicted eigenfunctions. (b)–(e) State comparisons for $n = 10$, 7, 3, and 0, respectively. Node counts are imposed; node positions and lobe amplitudes are learned.

five prominent nodes at the FEM positions. The rotated state, however, acquires two additional low-amplitude sign changes in the left tail, near $x \simeq -10.3$ and -7.4, so that its raw count of interior crossings is seven rather than $n = 5$. Variational mixing thus improves the state as a whole while violating its exact nodal identity in regions of negligible amplitude. This example is consistent with the state-dependent behavior shown in Fig. 2(b).

A controlled ablation isolates the contribution of the imposed structure. An independently trained reduced-physics variant shares the data split, FEM targets, JEPA representation path, parameter count, and training budget, but has no Sturm phase decoder, nodal or orthogonality losses, positive-gap energy parameterization, residual training, or Rayleigh–Ritz path. Its aggregate ID fidelity (0.983) is comparable to that of the direct route, but this aggregate conceals opposite behaviors. For the 667 test Hamiltonians outside the half-line and radial families, removing the structure lowers the mean fidelity from 0.990 to 0.980 (paired difference −0.0095, 95% CI $[-0.012, -0.008]$), doubles the fraction of states with $F < 0.9$, and triples the mean residual. The persistent nodal count is retained for only 81% of these states, almost always failing through spurious additional nodes, and for all 11 states in only 25% of these Hamiltonians, compared with 100% for the structured model. Without the positive-gap parameterization, the predicted energies violate spectral ordering in 22% of all ID Hamiltonians. Conversely, the unconstrained decoder reaches $F = 0.999$ on the 116 half-line and radial Hamiltonians, where the phase decoder reaches 0.916, identifying the decoder parameterization near the singular endpoint as the main limitation of the structured model; even there, the uncon-

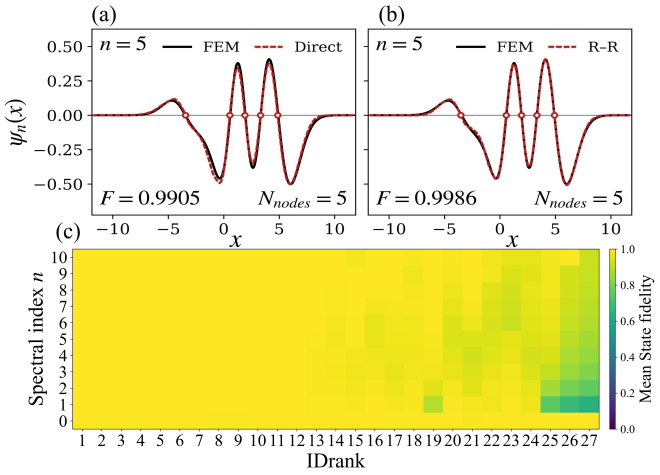


FIG. 4. State-resolved reconstruction and Hamiltonian dependence. (a) Direct prediction for an $n = 5$ state of an ID single-barrier Hamiltonian. (b) Rayleigh–Ritz state for the same Hamiltonian and spectral index, with improved fidelity and lower residual. Open circles and the displayed N_{nodes} mark prominent nodes, i.e., zeros separating two lobes whose peak amplitudes exceed 5% of the maximum; the Rayleigh–Ritz state additionally contains two raw sign changes in the low-amplitude left tail (seven raw crossings in total). (c) Mean direct fidelity as a function of spectral index and Hamiltonian-family identifier (IDrank) on the ID test set. The correspondence between IDrank and Hamiltonian family is listed in Table IV. Low-fidelity regions are concentrated in a few families and do not grow monotonically with the spectral index, showing that learning difficulty depends on the potential as well as on the state index.

strained energies are unreliable. High overlap therefore again does not imply structural admissibility or operator consistency (EM).

The sensitivity to variational rotation is also not explained simply by small spectral gaps. The association between adjacent spectral gaps and post-Ritz loss of resolved topology is weak ($r \simeq 0.14$), whereas the direct H^1 error correlates more strongly with subsequent topological loss ($r \simeq 0.53$). These diagnostics suggest that inaccurate trial-state shapes, rather than near-degeneracy alone, contribute to the larger corrections of the difficult intermediate states. The full gap and mixing diagnostics are reported in the EM.

Reconstruction difficulty depends jointly on spectral index and Hamiltonian family, as shown in Fig. 4(c). High-fidelity regions coexist with localized failures, and errors need not grow monotonically with the number of nodes. Transfer to unseen analytical functional forms makes this limitation more pronounced. The direct mean fidelity on that subset of 384 Hamiltonians is approximately 0.669, an average of near-perfect transfer to Scarf-II (0.999) and cosh wells (0.995) and a severe failure for Eckart potentials (0.013). Rayleigh–Ritz increases the subset average only to 0.673 and does not recover the inaccurate Eckart predictions (0.021). Structural admissibility can therefore survive a failure of the learned Hamiltonian dependence, and an inadequate trial subspace cannot generally be repaired by rotation within it.

The residual provides a practical diagnostic of these failures without requiring a reference eigensolution at inference. On the ID test set, the independent residual has a Spearman correlation of approximately 0.619 with infidelity. Ranking the test states by this residual alone selects more reliable predictions: mean fidelity rises from 0.9786 at full coverage to 0.9930 at 70% coverage. This is selective prediction, not an improvement of the discarded states. Likewise, a nodal safeguard can reject topology-violating refinements, but restores the direct output for approximately 96.3% of ID Hamiltonians rather than systematically improving its fidelity. Calibration and refinement diagnostics are given in the EM.

In summary, the surrogate separates structural admissibility from operator consistency and variational accuracy. Exact enforcement of the Sturm class removes a known spectral constraint from the learning task, but localization, node positions, and state identity still depend on the Hamiltonian. The predicted states can provide an accurate variational eigenspace even when their individual reconstructions are imperfect; conversely, Rayleigh–Ritz can improve that representation without guaranteeing preservation of state-resolved nodal identity. The reduced-physics ablation confirms this separation from the opposite direction: an unconstrained model of the same capacity can reach a comparable overlap while losing the nodal class, spectral ordering, and the diagnostic value of the residual. Structural admissibility, operator consistency, and variational quality therefore provide complementary, rather than interchangeable, validation criteria for learned eigenstates.

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END MATTER

Data contract and numerical reference

The numerical reference used to train and assess the surrogate was independently validated to ensure that the observed prediction errors and structural effects do not originate from the finite-element discretization. The FEM eigensolver was first benchmarked against analytically solvable systems. For the infinite square well and the harmonic oscillator, the converged eigenvalues reproduce the analytical spectra with maximum relative errors of approximately 2.4×10^{-5} and 3.3×10^{-5}, respectively. Halving the mesh spacing changes the eigenvalues by at most 2.5×10^{-4} in relative terms for every family except the Gaussian well (at most 1.2×10^{-3}), and the domain-convergence error remains below 5.4×10^{-4}.

TABLE II. Compact data contract for the dataset and numerical reference used throughout the results. Complete Hamiltonians are assigned to individual partitions, preventing different eigenstates of the same operator from appearing in different splits.

Quantity	Specification
Hamiltonians	5184
Potential families	27
Hamiltonians per family	192
Training / validation / test	3618 / 783 / 783
States per Hamiltonian	11 ($n = 0, \ldots, 10$)
Boundary conditions	Homogeneous Dirichlet
Kinetic coefficient	$\kappa = \hbar^2/(2m) = 0.5$
Reference discretization	P1 finite elements
FEM quadrature	Gauss–Legendre, order 4
Reference FEM grid	2049 nodes
Surrogate representation grid	513 nodes
Validation grid	1025 nodes
Spatial coordinate	$t = (x - x_{\mathrm{left}})/L$
Potential scaling	$U = [\min(V, V_{\mathrm{cap}}) - \mathrm{wmed}(V)]L^2/\kappa$
Energy scaling	$\epsilon = (E - E_{\mathrm{gauge}})L^2/\kappa$
Potential scale statistic	$U_{\mathrm{scale}} = 3643.21696$
Normalization statistics	Determined from training set only
Split unit	Complete Hamiltonian

The generalized FEM eigenstates were further tested directly in their native mass metric. For the coefficient matrix C of the computed eigenstates, the Gram matrix $C^\dagger MC$ reproduces the identity to numerical precision, with a maximum normalization error of 5.8×10^{-15} and a maximum Gram-matrix deviation of 1.4×10^{-14} across all families and parameter levels of the solver validation. All reference eigenfunctions also satisfy the expected Sturm node count. Operator consistency was checked independently by reassembling the weak Schrödinger operator with a higher quadrature order and evaluating

$$\mathbf{r}_n^{\mathrm{ind}} = H_{\mathrm{ind}}\mathbf{c}_n - E_n M_{\mathrm{ind}}\mathbf{c}_n.$$

Across the 81 family and parameter levels of the solver validation, the largest normalized independent residual of the reference states has a median of 2.2×10^{-5} and never exceeds 2.4×10^{-3}, showing that the numerical reference satisfies the eigenvalue problem well below the error scale of the learned surrogate (median direct residual 4.1×10^{-2}).

Finally, train, validation, and test partitions were constructed from complete Hamiltonians rather than individual eigenstates, preventing states from the same operator from appearing in different subsets. No Hamiltonian-ID overlap was detected between partitions, and the minimum cross-split operator distance, 3.9×10^{-4}, remained above the adopted near-duplicate threshold of 10^{-4}. Together, these tests establish that the FEM

reference solutions, their spectral structure, and the data partitions are numerically consistent, so that the discrepancies discussed in the main text arise from the learned operator-to-state approximation rather than from the reference solver.

Latent representation and training

The solution branch first projects FEM eigenfunctions onto a Dirichlet-compatible sine basis and encodes them in a trainable latent representation. Its reconstruction objective anchors this representation to reference eigenstates, including normalization, boundary behavior, nodal structure, and the weighted inner product. This solution autoencoder is specific to the present construction. The pretrained online encoder initializes a target encoder, which subsequently receives no direct gradient updates and follows the exponential moving average

$$\theta_{\mathrm{tgt}} \leftarrow \tau\theta_{\mathrm{tgt}} + (1-\tau)\theta_{\mathrm{on}}. \tag{5}$$

This online/target strategy follows I-JEPA and V-JEPA [16, 17]; the EMA is an optimization mechanism, not a physical constraint.

A context encoder receives the dimensionless potential, domain, boundary, grid, and geometrical information and produces z_V. A state-conditioned predictor maps z_V to $\widehat{z}_n$, while the target encoder maps the corresponding FEM state to z_n^{tgt}. The latent predictive objective is

$$\mathcal{L}_{\mathrm{JEPA}} = \left\| \widehat{z}_n - \mathrm{sg}(z_n^{\mathrm{tgt}}) \right\|_2^2, \tag{6}$$

where sg denotes stop-gradient. The squared L_2 objective follows I-JEPA and its molecular-graph extension [16, 18]. V-JEPA provides precedent for stop-gradient and EMA targets but uses an L_1 predictive objective [17]. AeroJEPA provides a related context-to-physical-field precedent, not a quantum-mechanical model, and uses a different latent regularization strategy [19].

Physical supervision is introduced progressively after latent alignment. Wavefunction reconstruction constrains spatial shape, while normalization, weighted overlaps, and the Schrödinger residual assess whether the outputs approximate a consistent eigenspace. Boundary conditions and nodal class are imposed by the decoder rather than established by these losses. This separation between quantities enforced by construction and quantities encouraged through the objective is closely related to the distinction between hard and soft constraints in physics-informed learning [20]. During inference the solution and target encoders are removed; the context encoder, predictor, and decoding branches use only operator-side information.

The quality of the learned solution manifold was assessed before evaluating the complete operator-to-state mapping. The pretrained solution autoencoder reconstructs the FEM eigensolutions with a mean fidelity of $\mathcal{F}_{\mathrm{AE}} = 0.9999989$, an aligned relative L^2 error of approximately 1.5×10^{-4}, and a relative H^1 error of 1.86×10^{-3}. Normalization is preserved to within 1.4×10^{-6}, the maximum weighted orthogonality error is approximately 3.1×10^{-6}, the Dirichlet boundary conditions are satisfied exactly, and the persistent-node accuracy is unity. The latent solution manifold is therefore substantially more accurate than the complete operator-to-state prediction, indicating that the principal difficulty lies in locating the appropriate eigensolution from the supplied Hamiltonian rather than in reconstructing a state from its latent representation.

Figure 5 summarizes the separation between learned Hamiltonian-dependent shape and imposed or explicitly tested physical structure. Orthogonality is tested in the mass metric and enforced by the multi-state projection, not guaranteed by the direct single-state decoder.

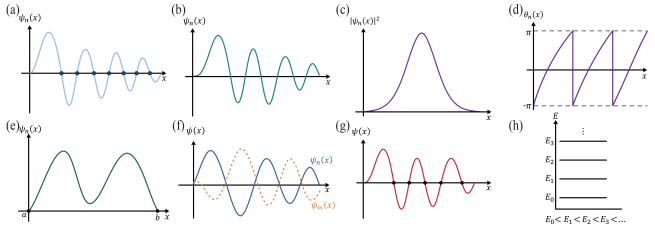


FIG. 5. Learned and imposed or tested structure in the predicted eigenstates. (a) Nodal geometry, represented by the spatial positions of the zeros of $\psi_n(x)$. (b) Relative lobe amplitudes of the eigenfunction. (c) Spatial localization, illustrated through the probability density $|\psi_n(x)|^2$. (d) Phase distribution $\theta_n(x)$ along the domain. (e) Homogeneous Dirichlet boundary conditions, $\psi_n(a) = \psi_n(b) = 0$. (f) Orthogonality between different states, $\langle\psi_n|\psi_m\rangle = 0$ for $n \neq m$. (g) Sturm nodal class, with the nth state containing exactly n interior nodes. (h) Spectral ordering of the eigenvalues, $E_0 < E_1 < E_2 < E_3 < \cdots$. Panels (a)–(d) correspond to Hamiltonian-dependent quantities learned from $V(x)$, whereas panels (e)–(h) summarize structural properties imposed or explicitly evaluated from the underlying physics.

Discrete operators and spectral constraints

For the decoder in Eq. (2), positivity is imposed by

$$q_n(t) = \mathrm{softplus}[\eta_n(t)] + \varepsilon > 0, \tag{7}$$

with $\varepsilon > 0$. Together with $A_n(t) > 0$, this yields

$$N_{\mathrm{nodes}}(\widehat{\psi}_n) = n. \tag{8}$$

This exact continuous count is distinct from the number of crossings resolved by a finite-grid numerical detection criterion.

Once the continuous prediction is represented in the same finite-element trial space used for the reference problem, it takes the form

$$\widehat{\psi}_{n,h}(x) = \sum_i \widehat{c}_{n,i} N_i(x), \qquad \widehat{\mathbf{c}}_n = \left(\widehat{c}_{n,1}, \ldots, \widehat{c}_{n,N}\right)^{\mathsf{T}}. \tag{9}$$

Thus $\widehat{\mathbf{c}}_n$ is not a separate network prediction, but the discrete coefficient vector representing the reconstructed state $\widehat{\psi}_n$.

The energy branch is constrained analogously at the spectral level. Rather than receiving FEM eigenvalues or predicting mutually independent energies, an operator-only auxiliary spectrum is constructed by projecting the dimensionless Hamiltonian onto a fixed K-dimensional Dirichlet sine basis,

$$H_{ij}^{(\sin)}[V] = \left\langle \phi_i \middle| \widehat{H}[V] \middle| \phi_j \right\rangle, \tag{10}$$

and diagonalizing the resulting reduced matrix. The corresponding auxiliary eigenvalues provide an operator-dependent spectral scale from which the neural energy branch predicts corrections to successive gaps. This construction depends only on the supplied operator and fixed basis functions: no FEM eigenvalues or eigenvectors enter the calculation, and the full finite-element generalized eigenproblem is not solved. Positive-gap parameterization guarantees

$$\widehat{E}_0 < \widehat{E}_1 < \cdots < \widehat{E}_{N-1}, \tag{11}$$

while the final numerical values of the energies remain determined by the learned correction. Thus the wavefunction decoder imposes the Sturm ordering in nodal space, whereas the energy branch imposes the corresponding ordering in the spectrum.

Individual topology-preserving states are not guaranteed to be mutually orthogonal. In the continuous quantum description, normalization and orthogonality are defined by

$$\langle \psi_n | \psi_m \rangle = \int_a^b \psi_n^*(x)\psi_m(x)\,\mathrm{d}x = \delta_{nm}. \tag{12}$$

In the finite-element representation, the same inner product and the Hamiltonian matrix element become

$$\begin{aligned} \langle \psi_n | \psi_m \rangle &\longrightarrow \mathbf{c}_n^\dagger \mathbf{M} \mathbf{c}_m, \\ \left\langle \psi_n \middle| \hat{H}[V] \middle| \psi_m \right\rangle &\longrightarrow \mathbf{c}_n^\dagger \mathbf{H} \mathbf{c}_m, \end{aligned} \tag{13}$$

where $M_{ij} = \int_a^b N_i^*(x) N_j(x)\,\mathrm{d}x$ is the finite-element mass (overlap) matrix, namely the discrete metric of the state space, and $H_{ij} = \langle N_i | \hat{H}[V] | N_j \rangle$ is the discrete Hamiltonian matrix. The matrices $\mathbf{H}$ and $\mathbf{M}$ are assembled directly from the supplied operator and finite-element basis functions, without requiring an eigensolution, and are used to evaluate normalization, orthogonality, and operator consistency.

The Gram matrix of direct predictions is

$$G_{nm} = \widehat{\mathbf{c}}_n^\dagger \mathbf{M} \widehat{\mathbf{c}}_m. \tag{14}$$

Weighted orthonormalization produces the basis $\mathbf{Q}$ used in Eq. (4). The reduced Rayleigh–Ritz calculation diagonalizes only the Hamiltonian within the predicted span, not the full FEM generalized eigenproblem. The weak residual in Eq. (3) likewise uses operator-assembled matrices without a reference eigensolution. Independent residual checks reassemble the weak operator with a higher quadrature order.

Reconstruction, ablation, and variational diagnostics

The direct-state fidelity distribution is strongly asymmetric. Its lower percentiles are $P_5 = 0.8688$, $P_{10} = 0.9314$, and $P_{25} = 0.9921$, whereas $P_{75} = 0.99965$ and $P_{95} = 0.99996$. The mean and median relative H^1 errors are approximately 0.100 and 0.040, respectively, indicating greater sensitivity of spatial derivatives than of the global overlap to reconstruction errors.

The energy estimates exhibit an even stronger improvement. Direct predictions have a median absolute energy error of approximately 5.3×10^{-2}, whereas the mean reaches approximately 3.1×10^2 because of a small number of extreme outliers. After the reduced Rayleigh–Ritz diagonalization, the median and mean errors decrease to approximately 6.1×10^{-4} and 1.6×10^{-2}, respectively.

To determine how the explicit physical structure contributes to predictive accuracy and structural reliability, a reduced-physics variant was trained with the same FEM solver, data generation, split, normalization, JEPA/EMA representation path, curriculum, training budget (4.09×10^4 versus 3.97×10^4 optimizer updates), seed, and total parameter count (1.70×10^6) as the complete model. The Schrödinger-residual loss and residual-guided refinement, the Sturm phase decoder and nodal losses, the positive-gap energy parameterization, the orthogonality loss, and the Rayleigh–Ritz training and inference path were removed. Dirichlet boundaries, normalization, and the supervised wavefunction, energy, H^1, density, tail, reflection, and spectral-subspace terms were retained. Residual, nodal, ordering, and Gram diagnostics were evaluated only after prediction. Both models were trained with a single seed.

Table III compares both models state by state on the identical ID test set. Outside the four half-line and radial families, the structured direct route is more accurate in fidelity, H^1 error, residual, and energy, and preserves the persistent nodal count for every state, whereas the reduced-physics model retains the raw nodal class for only 19% of the states. Of its persistent-count failures, 99.7% contain additional nodes. In the half-line and radial families the comparison reverses for the wavefunction overlap (0.999 versus 0.916); there, however, the unconstrained model has a mean residual of 0.98, a median energy error of 1.9×10^4, and ordering violations in 68% of the Hamiltonians, so that its accurate shapes are not accompanied by a consistent eigenpair. The residual is also a weaker confidence score without physics training: its Spearman correlation with infidelity is 0.14 (all ID states) instead of 0.62, and retaining the lowest-residual 70% of the states gives a mean fidelity of 0.983 instead of 0.993. Outside the training distribution the picture is similar. On the OOD suites other than the unseen functional forms, the mean fidelity is 0.968 instead of 0.981 and the persistent count is preserved for 81% instead of 99.6% of the states; on the unseen functional forms both models fail for Eckart potentials (0.108 and 0.013), giving subset averages of 0.700 and 0.669.

TABLE III. Reduced-physics ablation on the ID test set. The reduced-physics (RP) model and the complete model are trained on the identical split with the same budget; the direct and Rayleigh–Ritz (RR) routes belong to the complete model. Fractions refer to states unless stated otherwise; "all 11" is the fraction of Hamiltonians whose 11 states all have the correct persistent node count. Half-line and radial families: `Half_harmonic`, `Kratzer`, `Radial_coulomb`, and `Radial_oscillator`.

Metric	RP	Direct	RR
All ID Hamiltonians (783)			
Mean fidelity	0.983	0.979	0.994
Mean residual	0.323	0.108	0.092
Ordering violations (Hamiltonians)	22%	0%	0%
Spearman(residual, $1 - F$)	0.14	0.62	0.53
Excluding half-line and radial families (667)			
Mean fidelity	0.980	0.990	0.999
Fraction with $F < 0.9$	5.4%	2.6%	0.3%
Mean relative H^1 error	0.110	0.067	0.031
Mean residual	0.209	0.064	0.046
Median energy error	0.079	0.047	5×10^{-4}
Raw nodal class	19%	100%	71%
Persistent node count	81%	100%	93%
Persistent count, all 11	25%	100%	64%
Half-line and radial families (116)			
Mean fidelity	0.999	0.916	0.969
Mean residual	0.978	0.362	0.360
Persistent node count	60%	12%	49%

The state-dependent sensitivity to the variational rotation was examined in greater detail to determine whether the effect could be attributed to small spectral separations. Spectral gaps were defined as

$$\Delta E_n = E_{n+1}^{\mathrm{RR}} - E_n^{\mathrm{RR}}, \tag{15}$$

using the Ritz eigenvalues, whose median absolute deviation from the FEM reference (6.1×10^{-4}) is about 0.1% of the median nearest-neighbor gap. Loss of nodal identity is defined as failure of the Rayleigh–Ritz state to satisfy the resolved-topology criterion of Fig. 2(b).

No substantial association is observed between spectral separation and the post-Ritz loss of nodal identity. The Pearson correlation is only $r \simeq 0.14$ for the adjacent spectral gap and remains weak, $r \simeq 0.13$, when the nearest-neighbor gap is considered.

A substantially stronger relation is found with the quality of the direct trial states. The direct H^1 error correlates with the subsequent loss of nodal identity with $r \simeq 0.53$, while lower direct fidelity is associated with stronger variational mixing, measured by the displacement of each state under weighted orthonormalization, with $r \simeq -0.72$. The enhanced sensitivity of these states is therefore not primarily explained by small spectral gaps. Instead, states that enter the variational calculation with poorer spatial accuracy undergo larger corrections within the predicted subspace and are correspondingly more susceptible to changes in their state-resolved nodal identity.

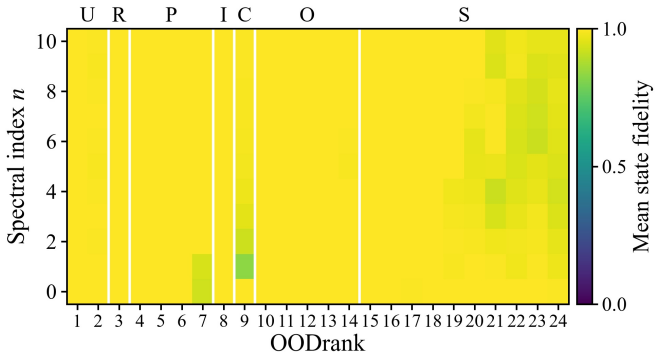


FIG. 6. Mean direct state fidelity as a function of spectral index and out-of-distribution suite (OODrank), for every evaluated suite except the Eckart potentials, whose fidelity (0.013 on average) would saturate the color scale. Letters and white lines mark the type of distribution shift: unseen functional form (U), reparameterization of a seen basis (R), parameter extrapolation (P), interpolation (I), near-continuum states (C), operator covariance or resolution change (O), and same-family challenge (S). The correspondence between OODrank and suite is listed in Table V.

In-distribution families

TABLE IV. In-distribution Hamiltonian families and their IDrank identifiers. Families are ranked by decreasing mean direct fidelity on the ID test set. Family names retain the dataset identifiers.

IDrank	Family	IDrank	Family
1	`Infinite_box`	15	`Multiwell`
2	`Poschl_h`	16	`Single_barrier`
3	`Harmonic`	17	`Multiple_barriers`
4	`Asymptotically_constant`	18	`Procedural_composite`
5	`Morse`	19	`Radial_coulomb`
6	`Anharmonic_polynomial`	20	`Harmonic_compact`
7	`Gaussian_well`	21	`Gaussian_barrier`
8	`Poschl_t`	22	`Fourier_random`
9	`Double_well`	23	`Localized_defect`
10	`Chebyshev_random`	24	`Gaussian_random_field`
11	`Asymmetric_well`	25	`Kratzer`
12	`Reflected_potential`	26	`Half_harmonic`
13	`Composite_confining`	27	`Radial_oscillator`
14	`Smooth_perturbed`		

Transfer is strongly dependent on the type of distribution shift. Figure 6 resolves the direct fidelity by spectral index for the evaluated out-of-distribution suites. Parameter extrapolation and interpolation, operator covariance and resolution changes, and structured perturbations of Hamiltonians already represented in the training distribution all retain high fidelity, with the lowest state-resolved values in the near-continuum suite (down to 0.83), followed by the controlled-smooth-perturbation and localized-defect suites (down to 0.92). Transfer to genuinely unseen analytical functional forms is considerably more heterogeneous. Across the unseen functional-form set (128 Hamiltonians for each of three generators that are never used for training or model selection), the direct mean fidelity is approximately 0.669. This average, however, com-

TABLE V. Out-of-distribution suites and their OODrank identifiers. Within each shift type, suites are ranked by decreasing mean direct fidelity. Suites 1–2 contain 128 Hamiltonians each and suite 3 contains 16; suites 15, 17, 20, 22, 23, and 24 are the untouched 29-Hamiltonian slices of the ID test set; all other suites are single diagnostic Hamiltonians.

OODrank	Shift	Family	OODrank	Shift	Family
1	U	Scarf_ii	13	O	Morse
2	U	Cosh_well	14	O	Harmonic
3	R	Rosen_morse_ii	15	S	Anharmonic_polynomial
4	P	Harmonic	16	S	Asymmetric_well
5	P	Morse	17	S	Double_well
6	P	Fourier_random	18	S	Multiwell
7	P	Double_well	19	S	Multiple_barriers
8	I	Harmonic	20	S	Single_barrier
9	C	Radial_coulomb	21	S	Smooth_perturbed
10	O	Harmonic	22	S	Fourier_random
11	O	Harmonic	23	S	Localized_defect
12	O	Morse	24	S	Gaussian_random_field

bines markedly different regimes. Scarf-II potentials retain a mean fidelity of approximately 0.999 and the cosh-well family approximately 0.995, whereas Eckart potentials fall to approximately 0.013. Reduced Rayleigh–Ritz projection increases the aggregate fidelity of this unseen subset only to approximately 0.673, and does not recover the severely inaccurate Eckart predictions (0.021). The direct outputs nevertheless retain the persistent nodal count for all unseen states, so structural admissibility survives even where the learned Hamiltonian dependence fails.

Refinement and calibrated confidence

The weak residual supplies a physics-derived refinement direction whose magnitude is modulated by a trainable correction module. Because orthonormalization, variational rotation, and residual refinement need not preserve the prescribed nodal identity, a final safeguard accepts a corrected state only when that identity survives; otherwise it restores the direct prediction.

Residual-based refinement provides a complementary mechanism for improving states that are already close to an eigensolution. In favorable cases, successive correction steps progressively reduce the Schrödinger residual. For the infinite-box benchmark, for example, the independent residual decreases from approximately 9.1×10^{-3} without refinement to 3.2×10^{-3} after five iterations. The improvement is not universal, however. On the unseen functional forms, no proposed correction is accepted, and the outputs remain unchanged; on the ID test set, 0.37 steps are accepted per Hamiltonian on average. This indicates that local residual-based refinement cannot systematically recover a qualitatively incorrect initial state.

The hybrid route consequently operates primarily as a conservative safeguard. A refined or variationally rotated state is retained only when the prescribed nodal identity survives the correction; otherwise, the direct topology-preserving state is restored. Approximately 96.3% of the ID Hamiltonians activate this fallback mechanism. The hybrid procedure therefore provides protection against topology-violating post-processing rather than a systematic increase in direct reconstruction fidelity.

The pronounced low-fidelity tail raises the question of whether inaccurate predictions can be identified without access to the reference eigensolution. For this purpose, the independent Schrödinger residual was used as an operator-derived confidence score. The residual is available directly from the predicted pair $(\widehat{E}_n, \widehat{\psi}_n)$ and the supplied Hamiltonian, and therefore requires no FEM eigenvalue or eigenfunction at inference. On the ID test set, the residual exhibits a substantial monotonic association with the state infidelity, with a Spearman correlation of approximately 0.619.

To quantify this relation, the direct test states were ranked by their independent residual and the lowest-residual fraction was retained. Because selection depends only on this ranking, any monotonic calibration of the residual against infidelity would select the same states. With infidelity as the risk, the area under the risk–coverage curve is approximately 0.0061. Selective prediction shows that the cases with higher residual account for a substantial fraction of the failure tail. The mean fidelity increases from 0.9786 when all test states are retained to 0.9802, 0.9869, and 0.9930 at 90%, 80%, and 70% coverage, respectively, reaching 0.9977 when the lowest-residual 50% of the predictions are retained. Operator consistency therefore provides not only a post hoc diagnostic but also a target-free criterion for identifying predictions that are more likely to deviate from the corresponding eigensolution. No reference eigensolution is required to evaluate the residual or apply the ranking at inference.